

CANADIAN ARCTIC

A far northern place of ice and snow

The Canadian Arctic includes one of the world's largest archipelagos—36,563 islands, most of which are uninhabited by people. The easternmost islands are mountainous, becoming lower lying in the west. For much of the long, dark winter, land and sea are bound in a vast ice-scape, broken only by rocky island peaks and occasional polynyas—areas of sea that freeze late and thaw early. Polynyas are a vital resource for marine mammals including belugas and bowhead whales, which use them as breathing holes, and for seals and polar bears, which need to access the water from the sea ice. In summer, strong tides sweep the channels between the islands.

Frozen ground

On land, the top three feet of soil thaws briefly in summer, but the ground beneath is permanently frozen (permafrost). Plant life is limited to mosses, lichens, and around 200 species of grass, sedge, hardy forbs, and dwarf shrubs. Land mammals able to withstand the cold include caribou, muskoxen, Arctic foxes, and lemmings. The number of invertebrate species is low, but mites and springtails become superabundant in summer, providing food for breeding migrant birds such as Arctic terns, ivory gulls, common eiders, and red phalaropes.

Melting of the permafrost releases methane and carbon, increasing the rate of global warming



ARCTIC TERN

LONG-DISTANCE TRAVELER

The Arctic tern travels further than any animal on the planet as it migrates between its Arctic breeding areas and the Antarctic. Many Arctic animals head south to avoid the winter—caribou make a shorter but equally arduous trip to the tundra regions of Canada.

HOODED SEAL MOTHER AND PUP

TOUGH START IN LIFE

Newborn hooded seals feed on super-rich milk that enables them to increase their weight from 55-100 lb (25 to 45 kg) in just four days. After that, their mothers return to the sea to mate and feed. Each pup must learn to swim, dive, and hunt before it starves.

WOOD FROG

WINTER FREEZE

Wood frogs pass the winter dormant, their blood and skin frozen solid. To survive, they pump their cells full of a protective glucose syrup made by the liver. In contrast, Arctic cod have antifreeze proteins that prevent their blood freezing.

Of 36 mammal species here, 17 are marine

The Sun never sets in summer

Newborn hooded seals feed

WOOD FROG

WINTER FREEZE